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Bisphenol F suppresses insulin-stimulated glucose metabolism in adipocytes by inhibiting IRS-1/PI3K/AKT pathway

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A R T I C L E I N F O

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A B S T R A C T

Obesity is one of the risk factors of metabolic diseases. Decreased sensitivity to insulin or impairment of the insulin signaling pathway may affect the metabolism of adipose tissue. Bisphenol F (BPF) has been widely used in various products as a substitute for bisphenol A (BPA). BPA has been defined as "obesogen". However, knowledge about the correlation between BPF and obesity is very limited. This study was aimed to explore the effects of BPF on glucose metabolism and insulin sensitivity in mammalian tissues, using a mouse 3T3-L1 adipocyte line as the model. Differentiated 3T3-L1 adipocytes were treated with BPF at various concentrations for 24 h or 48 h, followed by the measurement of cell viability, lipid accumulation, expression levels of adipocytokines, glucose consumption, and impairment of the insulin signaling pathway. The results indicated that BPF had no effect on the size of 3T3-L1 adipocytes, but the expression of leptin, adiponectin and apelin was decreased, while that of chemerin and resistin was increased after 48 h of BPF treatment. Moreover, BPF inhibited the glucose consumption, the expression of GLUT4, and its translocation to the plasma membranes in 3T3-L1 adipocytes. Western blot analysis indicated that the activation of IRS-1/PI3K/AKT signaling pathway was inhibited by BPF, which resulted in reduced GLUT4 translocation. In conclusion, our data suggest that exposure of adipocytes to BPF may alter the expression of calorie metabolism-related adipokines and suppress insulin-stimulated glucose metabolism by impairing the insulin signaling (IRS-1/PI3K/AKT) pathway.

1. Introduction

Bisphenol F (BPF), chemically named 4,4-dihydroxydiphenylmethane, is a substitute for bisphenol A (BPA). As BPA has been shown to induce adverse effects on human health, especially through the activation of endocrine pathways, the use of BPA has been restricted (den Braver-Sewradj et al., 2020). For this reason, BPF is widely used in the manufacture of polycarbonate plastics, epoxy resins, structural adhesives, water pipes and food contact materials (Qiu et al., 2018; Wang et al., 2014). Recently, BPF was found in human plasma at concentrations three times higher than BPA (Usman et al., 2019). BPF was detected in 55% of the urine samples from residents in south China collected between 2009 and 2012, with a median concentration of 0.08 mg/L; similarly, it was found in about 60% of the urine samples from U. S. adults, with concentrations ranging from 0.15 to 0.54 μg/L (Yang et al., 2014; Ye et al., 2015; Zhou et al., 2014). Even though BPF is already largely used in our daily life, the potential adverse health effects of BPF remain still unclear, and it is important to study the potential harm of BPF to health.

Obesity is widely considered the result of disequilibrium between energy intake and expenditure. Although well-directed public health policies and individual treatment efforts have been made to combat the obesity epidemic, more than two billion people around the world are overweight or obese. In addition to central nervous system circuits as

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well as fuel turnover and metabolism, adipose tissue homeostasis is very important for understanding excessive weight gain and related comorbidities (González-Muniesa et al., 2017). Current evidence suggests that some environmental chemicals may interfere with energy metabolism and endocrine regulation in the adipose tissue (Darbre, 2017). BPA is a lipophilic compound that can accumulate in fat, present at detectable levels in 50% of breast adipose tissue samples from women (Fernandez et al., 2007). Moreover, BPA has been defined as “obesogen” and may interfere with cellular energy metabolism that results in its dysregulation (Alonso-Magdalena et al., 2010). The prolonged exposure to obesogens may be related to increased susceptibility to metabolic diseases (Muscogiuri et al., 2017). Considering that the structures of BPF and BPA are similar, BPF may has potential impact on adipocytes.

Adipose tissue is composed of mature adipocytes and a stromal vascular fraction (Han et al., 2015). As a major metabolic and energy-storing tissue, adipose tissue plays an important role in many metabolic pathways, including insulin signaling and insulin sensitivity (Yaribeygi et al., 2019). Adipocytes secrete a variety of adipokines, such as leptin, adiponectin, resistin, apelin and chemerin, which cause pleiotropic effects, including modulation of whole-body homeostasis, metabolism, and inflammation (Morris and Edwards, 2018; Szydło et al., 2016). Adipokines have protective properties when they are secreted at normal levels, however, the imbalance in levels of these adipokines is thought to promote the development of obesity-linked complications (Pham and Park, 2020). Many adipokines have been shown to affect the function of pancreatic β cells through regulating insulin release or influencing cell survival. In turn, insulin regulates lipolysis and promotes glucose uptake and lipid storage in adipocytes and other types of cells (Cantley, 2014). In adipose tissue, insulin regulates the energy and lipid metabolism via the activation of the insulin receptor (INSR), insulin receptor substrate (IRS) proteins, phosphoinositol 3-kinase (PI3K) and protein kinase B (also known as AKT) (Cignarelli et al., 2019). The PI3K/AKT signaling pathway then regulates the translocation of the insulin-sensitive glucose transporter type 4 (GLUT4) to the membrane of muscle and adipocytes for glucose uptake (Saltiel, 2021). GLUT4 controls glucose transport into adipose and muscle tissues in response to insulin (Rea and James, 1997). The amount of GLUT4 in adipose tissue represents a good marker of systemic insulin sensitivity (Graham et al., 2006).

Hence, in the present research, 3T3-L1 adipocytes were selected as an in vitro model to clarify the potential effects of BPF on obesity. The changes of lipid accumulation, glucose consumption and insulin function of 3T3-L1 adipocytes were detected and the potential mechanism was explored. It is hoped that our research could provide an experimental basis for the correlation between BPF and obesity.

2. Methods

2.1. Cell culture and differentiation

Cell culture. 3T3-L1 mouse preadipocytes was a kind gift from Dr. Changqing Zuo (Guangdong Medical University, Dongguan, China). 3T3-L1 preadipocytes were cultured in high glucose DMEM (Gibco Life Technologies, USA) containing 10% newborn calf serum (CS, Gibco Life Technologies, USA) and 1% nonessential amino acids (Gibco Life Technologies, USA), and the incubator conditions were maintained at 37 ℃ and 5% CO_2.

Cell differentiation. 3T3-L1 cells were seeded in 6-well plates at a density of 8×10^4 cells per well. After cell confluence, cells were cultured under the above conditions for another 48 h, and then the adipogenic differentiation was induced (Day 0). First, the cells were induced in differentiation medium I (DM I) including 0.5 nm 3-isobutyl-1-methylxanthine (IBMX, Sigma-Aldrich, USA), 1 μM dexamethasone (Sigma-Aldrich, USA), 10 μg/mL insulin (Solarbio, Beijing, China) and DMEM containing 10% FBS (Bovogen, New Zealand) (Zhao et al., 2019) for 72 h. Then the differentiation medium II (DM II) including 10 μg/mL insulin and DMEM containing 10% FBS was added and cultured for 48 h. Finally, the medium was replaced with the differentiation medium III (DM III) including DMEM containing 10% FBS for 4 days, during which the culture medium was changed every 2 days. The differentiation results were determined by microscopic observation.

2.2. BPF treatments

Treatments were done once the adipocytes were differentiated to study the effects of BPF on the glucose metabolism and insulin sensitivity of mature adipocytes. On Day 9 of differentiation, adipocytes were exposed to 0, 1, 5, 10, or 20 μM of BPF (≥ 98% pure, Sigma-Aldrich, USA), which was dissolved in 0.1% dimethyl sulfoxide (DMSO; Sigma-Aldrich, USA) for 24 h or 48 h. These concentrations were chosen according to the cell counting kit-8 (CCK-8, Dojindo Laboratories, Japan) assay. Untreated cells (0 μM BPF) were exposed to 0.1% DMSO (Fig. 1).

2.3. Cell viability assay

Cell viability was evaluated by using CCK-8 assay according to the manufacturer’s instructions. In brief, 3T3-L1 preadipocytes were seeded in 96-well plates at a density of 2×10^3 cells per well and differentiated to be mature adipocytes, then treated with BPF (0, 1, 5, 10, 20, 50, 100 μM) for 24 h or 48 h. The culture medium was replaced with 100 μL fresh medium containing 10 μL CCK-8 solution, and incubated at 37 ℃ for 1.5 h. Finally, the OD value was measured at 450 nm by a microplate reader (ELx808, BioTek, Instruments, Inc., Winooski, VT, USA) and the relative cell viability was presented as percentage of control.

2.4. Oil-red-O staining assay

After 48 h treatment of BPF on the 3T3-L1 adipocytes, the lipid accumulation of mature adipocytes was measured via an Oil-Red-O staining kit (Solarbio, Beijing, China) following the instructions of the manufacturer. Then, cells were visualized under a light microscope (EVOS XL Core; Invitrogen, Carlsbad, CA, USA) and photographed (Nikon 7100D; Nikon, Tokyo, Japan). Relative differences between 0 μM (DMSO) and BPF groups were measured by comparing the absorbance (490 nm) of Oil-Red-O eluates from the monolayers, which was obtained with 100% isopropanol and measured using microplate reader.

2.5. Intracellular triglyceride and total cholesterol quantification

After 48 h treatment of BPF, 3T3-L1 adipocytes were harvested after washing twice with phosphate-buffered saline (PBS). The amount of triglyceride (TG) and total cholesterol (TC) in the samples was measured with commercial assay kits (Nanjing Jiancheng Biotechnology, China) and the protein content was measured with BCA Protein Assay kit (Thermo Fisher Scientific, USA) following manufacturer’s instructions. The TG and TC content was normalized with protein concentration.

2.6. RNA isolation and qRT-PCR

The total RNA of the 3T3-L1 adipocytes was extracted with Trizol reagent (Thermo Fisher Scientific, USA), and then cDNAs were synthesized by reverse transcription kit (Takara Bio, Japan) according to the manufacturer’s instructions. The TB Green Real-time PCR kit was used and detected by Piko real 96 fluorescence quantitative PCR. qRT-PCR amplification was carried out under the following conditions for 40 cycles: 95 ℃ for 30 s, 95 ℃ for 15 s, 60 ℃ for 30 s. The primers of *tnf-α, il-6, mcp-1, leptin, adiponectin, apelin, chemerin, resistin and β-actin* were synthesized by Shanghai Generay Biotech Co, Ltd (Shanghai, China), which were listed in Table 1. The relative quantity of mRNA was calculated using the standard $2^{-\triangle\triangle CT}$ method with β-actin serving as an internal standard.

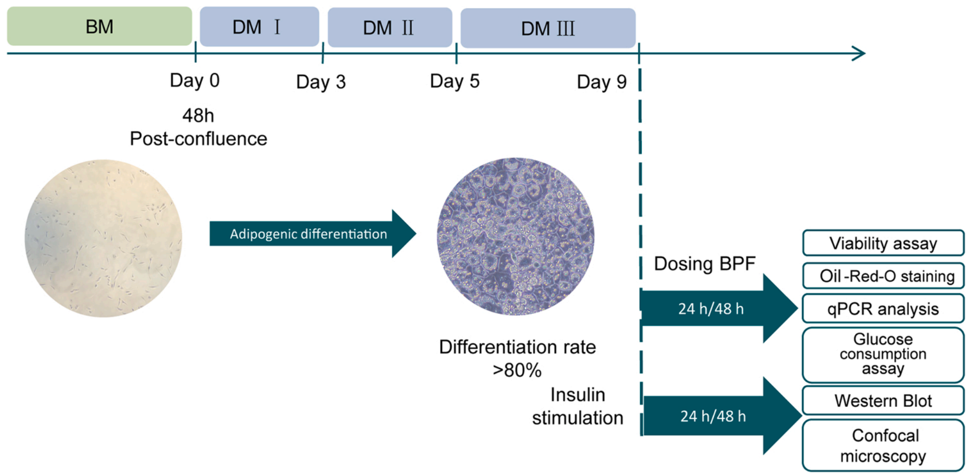

Fig. 1. Experimental protocol. BM: basal medium; DM I: differentiation medium I; DM II : differentiation medium II; DM III: differentiation medium III.

Table 1
Primer sequences used for quantitative real time PCR analysis.

Gene	Forward	Reverse
tnf-α	TGAGGTCAATCTGCCCAAGTA	AGGTCACTGTCCCAGCATCT
il-6	ACAGAAGGAGTGGCTAAGGA	AGGCATAACGCACTAGGTTT
mcp-1	TTCCTCCACCACCATGCAG	CCAGCCGGCAACTGTGA
leptin	GAGACCCCTGTGTCGGTTC	CTGCGTGTGTGAAATGTCATTG
resistin	AGACTGCTGTGCCTTCTGGG	CCCTCCTTTTCCTTTTCTTCCTTG
adiponectin	TGTTCCTCTTAATCCTGCCCA	CCAACCTGCACAAGTTCCCTT
chemerin	GACCAACTGCCCCAAGAA	GTCCATTTTAATGCAGGCCAG
apelin	CGAGTTGCAGCATGAATCTGAG	TGTTCCATCTGGAGGCAACATC
gLUT4	GTTGCGGATGCTATGGGT	GTCCGGCCTCTGGTTTC
β-actin	TGAGAGGGAAATCGTGCGTGAC	GCTCGTTGCCAATAGTGATGACC

2.7. Measurement of glucose consumption

The glucose consumption of the 3T3-L1 adipocytes was measured as described (Rivera Diaz et al., 2020). In brief, after treating with BPF for 24 h or 48 h, the 3T3-L1 adipocytes were fasted in DMEM with no glucose for 1 h, and then placed in high glucose DMEM (initial glucose concentration is 4.5 g/L) either in the absence (basal consumption) or in the presence of 100 nM insulin (stimulated consumption). 2 h later, the cell culture medium was collected to measure the basal and insulin-stimulated glucose consumption. Glucose levels in cell culture supernatant were measured with a glucose assay kit (Nanjing Jiancheng Biotechnology, China) according to the manufacturer's instructions. The absorbance at 505 nm was read in a microplate reader. Glucose consumption was calculated as the difference between initial glucose (IG) and extracellular glucose (EG) according to the equation: glucose consumption levels = (IG of the treatment group − EG of the treatment group) / (IG of the control group–EG of the control group) × 100%.

2.8. Western blot

After treating with different concentrations of BPF for 48 h, 100 nM insulin was added for 15 min, and then the total protein of the 3T3-L1 adipocytes in different groups was extracted respectively while the membrane protein was obtained using the Membrane and Cytosol Protein Extraction Kits (Beyotime, China). The protein phosphatase inhibitor cocktail tablet (Applygen, China) was added to the cell lysate following manufacturer's protocol during the protein extraction process to avoid dephosphorylation of phosphorylated proteins. BCA Protein Assay kit was used to measured protein concentration. Equal amount of protein was loaded in the SAS-PAGE gel (ComWin Biotech, China) to electrophorese and then transferred onto a PVDF membrane (Roche Appled Science, Germany). Following blocked with 5% milk (CST, USA) at room temperature for 1.5 h, the membrane was incubated with primary antibody like anti-p-IRS1, anti-p-PI3K, anti-p-AKT, anti-IRS1, anti-PI3K, anti-AKT, anti-β-actin (CST, USA) and anti- GLUT4 (Abcam, UK) at 4 ℃ overnight. After washing with TBST, the membranes were incubated with labeled secondary antibody at room temperature for 1.5 h. Finally, myECL™ Imager (Thermo Fisher Scientific, USA) was used for chemiluminescence detection, and the image was subjected to grayscale scanning analysis using Alpha View SA software after exposure and development. Densities of non-saturated bands were quantified by Image J software. The level of β-actin was used as control and measured on the same gel.

2.9. Immunofluorescence and confocal microscopy

The 3T3-L1 preadipocytes were differentiated and treated with BPF (20 μM) for 48 h on cell slides in the 6-well plate. Before being fixed with ice-cold methanol for 10 min, the cells were stimulated with insulin for 30 min. The cells were then washed and blocked with 5% BSA in PBS for 1 h at room temperature, followed by overnight incubation with the GLUT4 antibody (1:500 dilution in 5% BSA) at 4 ℃. After 1 h incubation of Alexa Fluor 488-labeled (Abcam, UK) secondary antibodies at room temperature, the cells were treated with mounting medium containing DAPI (Sigma-Aldrich, USA). Finally, the cells were observed and photographed by a confocal microscope (Leica TCS SP8 high-speed imaging system).

2.10. Statistical analyses

Statistical analysis was conducted with GraphPad PRISM 8.0 statistical software (San Diego, California, USA). All data was presented as mean ± SD and each experiment was performed at least three times. Statistical significance was assessed by One-way analysis of variance (ANOVA) with Duncan's comparison test to determine the significance of differences among groups, and a P value less than 0.05 was considered statistically significant.

3. Results

3.1. No effect of BPF on the lipid accumulation/viability of 3T3-L1 adipocytes

To determine the cytotoxicity of BPF on the 3T3-L1 adipocytes, CCK-8 assay was performed on the 3T3-L1 adipocytes after 24 h or 48 h treatments with BPF. At none of the test concentrations BPF was cytotoxic (Fig. S1). According to previous studies (Martínez et al., 2020; Ramskov Tetzlaff et al., 2020) and the results of CCK-8 assays in this study, concentrations of 1 μM, 5 μM, 10 μM and 20 μM BPF were selected for following experiments.

White adipose tissue expands in two ways. One is to enlarge the size of mature adipocytes by filling the cells with lipids, the other is to increase the number of adipocytes by promoting the differentiation of preadipocytes (Joe et al., 2009; Shin et al., 2020; Wang et al., 2013). Adipocytes can be used as a measure of obesity according to their size (hypertrophic obesity) and number (hyperplastic obesity) (Wang et al., 2013). In order to investigate whether BPF induced enlargement of mature adipocytes size by increasing the lipid accumulation, the Oil-Red-O staining assay , TG and TC contents were used to detect the lipid accumulation after 48 h of BPF treatment in 3T3-L1 adipocytes. The results showed that exposure to BPF for 48 h did not cause the changes of TG and TC contents in 3T3-L1 adipocytes compared with the control group, which was consistent with the results of Oil-Red-O staining assay (Fig. 2). These results showed no change in the content of lipids in 3T3-L1 adipocytes, suggesting that BPF may not increase lipid accumulation nor promote the enlargement of 3T3-L1 adipocytes size.

3.2. BPF influenced the expression of adipokines in 3T3-L1 adipocytes

Adipocytes are capable of synthesizing and releasing physiologically active molecules, including leptin, adiponectin, resistin, and chemerin, as well as inflammatory cytokines such as interleukin-6 (IL-6) and tumor necrosis factor (TNF-α). These secretory products of adipocytes are named adipokines (Xie and Chen, 2019). Adipokines can alter the metabolism of adipose tissue and even the whole organism body. Therefore, changes of adipokines can be used as an indicator of adipose tissue dysfunction and pathogenic diseases (Kim and Moustaid-Moussa, 2000; Oh et al., 2016). For investigating the effect of BPF exposure on the secreting function of 3T3-L1 adipocytes, the expression of adipokines including proinflammatory cytokines were measured following 24 h and 48 h treatments. After 24 h BPF treatment, there was a reduction in the mRNA expression of the *adiponectin* (20 μM), *resistin* (5 μM and 20 μM) and *apelin* (5 μM and 10 μM), while the proinflammatory cytokine *tnf-α*, *il-6* and *mcp-1* were not affected by BPF (Fig. 3A). Exposure to BPF for 48 h reduced the transcription of the adipokine *adiponectin*, *leptin* and *apelin*, while increased that of the *resistin* and *chemerin*. The proinflammatory cytokine *il-6* and *mcp-1* were decreased significantly by BPF at the concentrations of 1 μM and 20 μM (Fig. 3B). These results indicated that BPF may change the expression of adipokines in the 3T3-L1 adipocytes.

3.3. Suppression of glucose metabolism by BPF in 3T3-L1 adipocytes

The glucose balance in adipocytes may attribute to several mechanisms such as that by adipokines, which have various effects on glucose homeostasis (Rosen and Spiegelman, 2006). Adipokines are primarily secreted by the functional differentiated adipocytes (Ali et al., 2013).

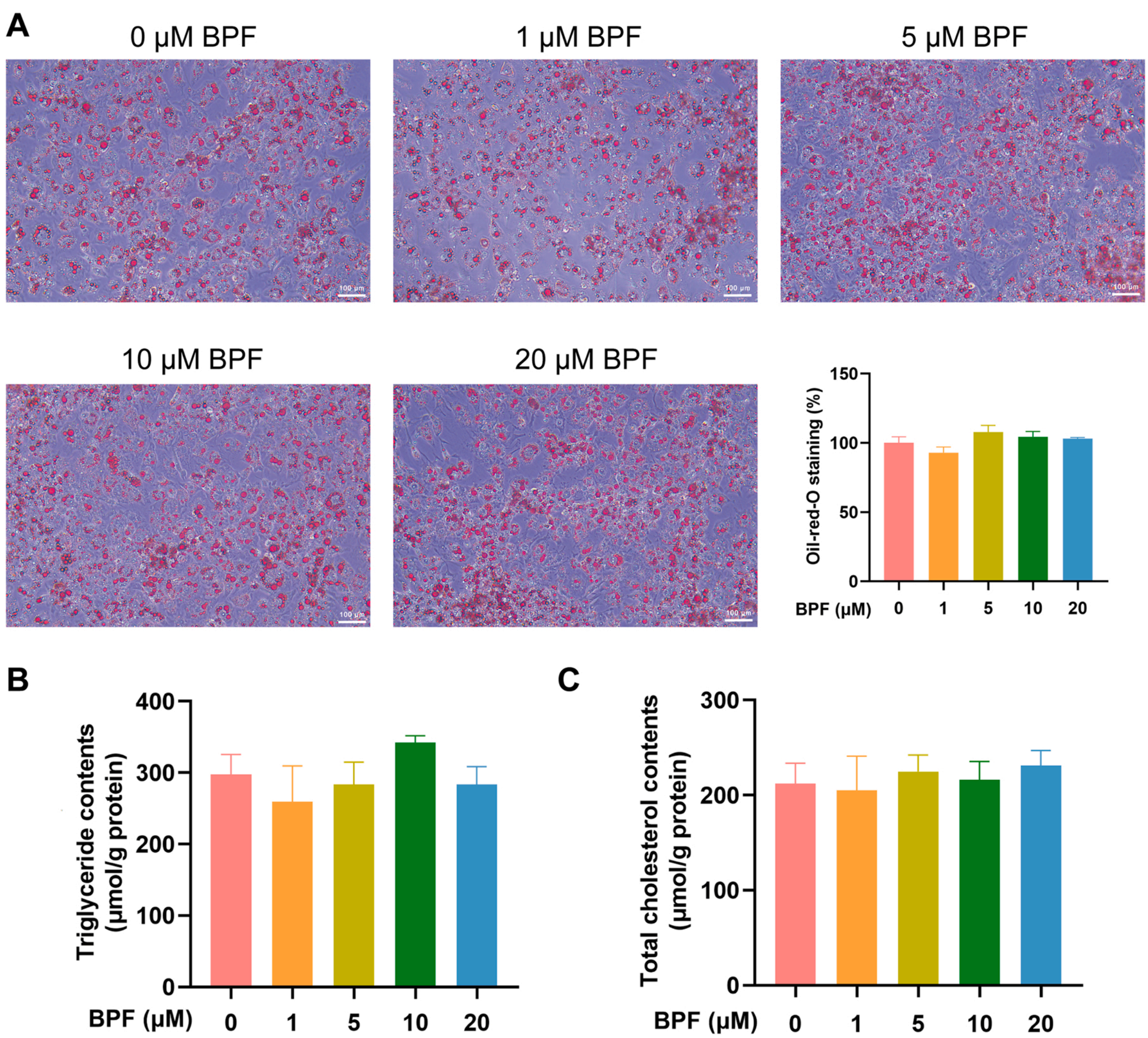

Fig. 2. BPF may not increase lipid accumulation in 3T3-L1 adipocytes. Oil red O staining assay (A), analyses of TG (B) and TC (C) was performed on the 3T3-L1 adipocytes after 48 h treatments of BPF. Data are presented as mean ± SD (n = 5).

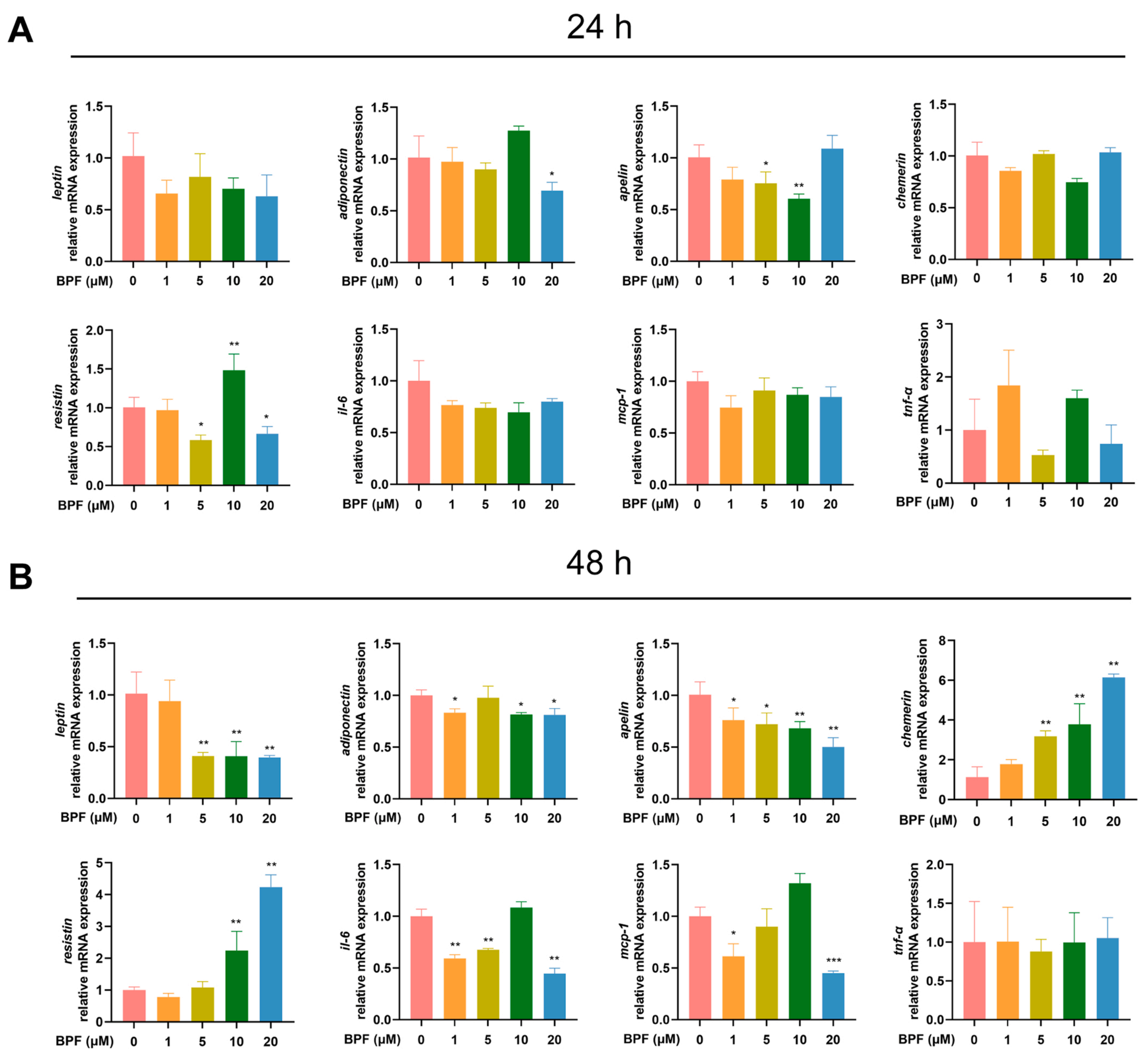

Fig. 3. BPF regulates the mRNA expression of adipokines in 3T3-L1 adipocytes after 24 h or 48 h treatment. After treating with BPF for 24 h (A) or 48 h (B), the relative mRNA expression of adipokine (*leptin, adiponectin, apelin, chemerin, resistin*) and proinflammatory cytokine (*tnf-α, il-6 and mcp-1*) were determined by qRT-PCR. Data are presented as mean ± SD (n = 3–6). * $p < 0.05$ vs. 0, ** $p < 0.01$ vs. 0, *** $p < 0.001$ vs. 0.

According to previous reports, leptin and adiponectin may exert anti-diabetic action while resistin tends to raise blood glucose (Rosen and Spiegelman, 2006).

As BPF could alter the expression of adipokines, including leptin, adiponectin, and resistin etc., we supposed that BPF might have an impact on the glucose metabolism in 3T3-L1 adipocytes. Therefore, we performed further experiments to verify this assumption. In the absence of insulin stimulation, the basal glucose consumption of 3T3-L1 adipocytes was detected when exposed to BPF for 24 h or 48 h. The results showed that the basal glucose consumption of 3T3-L1 adipocytes was decreased slightly by BPF (Fig. 4A and C). Furthermore, the glucose consumption of 3T3-L1 adipocytes was detected again after 24 h or 48 h of BPF exposure in the presence of insulin stimulation. The results showed that 24 h exposure with BPF decreased the glucose consumption slightly in 3T3-L1 adipocytes (Fig. 4B), which is similar to the decline of basal glucose consumption. Interestingly, the glucose consumption with insulin stimulation was significantly decreased by BPF after exposure for 48 h (Fig. 4D). These results suggested that BPF suppressed glucose metabolism, especially that stimulated by insulin in adipocytes after prolonged exposure.

3.4. Suppression of insulin-stimulated glucose metabolism by BPF in 3T3-L1 adipocytes via reduced expression and translocation of GLUT4

Glucose transporter type 4 (GLUT4) controls glucose transport into adipose and muscle tissues in response to insulin, which promotes its translocation from the cytoplasm to the plasma membrane (Bryant et al., 2002). If the insulin-stimulated glucose transport into adipocytes and muscle is impaired, it will cause insulin resistant and type 2 diabetes (Klip et al., 2019). Considering the importance of GLUT4 for glucose transport, we investigated whether BPF suppresses insulin-stimulated glucose consumption though acting on GLUT4 in 3T3-L1 adipocytes. Firstly, we analyzed the expression of GLUT4 in insulin-treated 3T3-L1 adipocytes and the effect of BPF after 48 h of exposure, by using

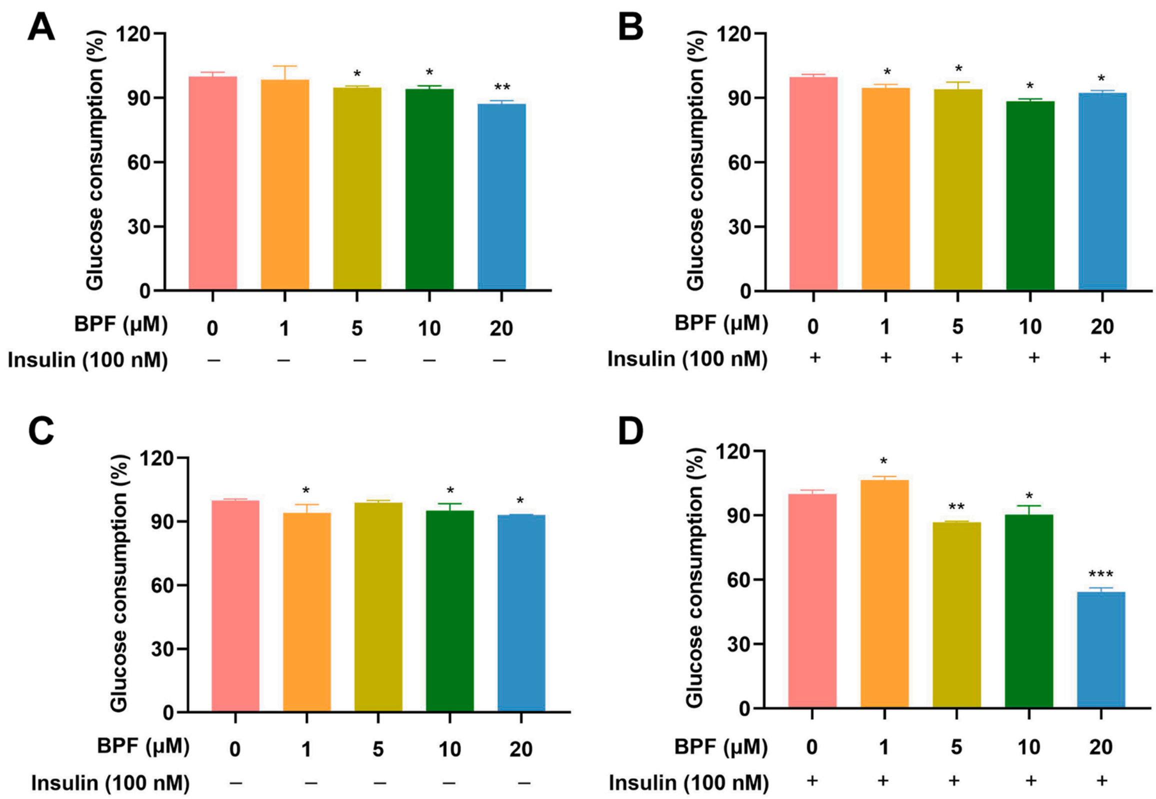

Fig. 4. BPF suppressed glucose metabolism of the 3T3-L1 adipocytes. After 24 h treatment (A and B) or 48 h treatment (C and D), BPF reduced the glucose consumption of 3T3-L1 adipocytes in absence (A and C) or presence (B and D) of insulin (100 nM). Data are presented as mean ± SD (n = 3–5). * $p < 0.05$ vs. 0, ** $p < 0.01$ vs. 0, ***$p < 0.001$ vs. 0.

qRT-PCR and western blot assays. As shown in Fig. 5, the transcription of GLUT4 was decreased significantly by BPF at 1 μM and higher concentrations, while the level of its protein expression was reduced only at the concentration of 20 μM (Fig. 5A and B). Next, we investigated the translocation of GLUT4 to the membrane of adipocytes using confocal microscopy. Immunofluorescence of GLUT4 showed that the fluorescence intensity of GLUT4 was significantly reduced in insulin-stimulated 3T3-L1 exposed to 20 μM of BPF exposure as compared with cells treated with only insulin (Fig. 5C and D), which was in accordance with the results of Western blot. These results suggested that BPF inhibited the expression and translocation of GLUT4 to the membrane that stimulated by insulin.

3.5. *Inhibition of BPF on the IRS-1/PI3K/AKT pathway in insulin-treated 3T3-L1 adipocytes*

Insulin-stimulated glucose uptake in adipocytes is heavily dependent on the activation of IRS-1-PI3K-AKT pathway (Petersen and Shulman, 2018), which can induce the translocation of GLUT4 from the cytoplasm to the plasma membrane (Jang et al., 2020). To elucidate the mechanism of BPF suppressing insulin-stimulated glucose metabolism in 3T3-L1 adipocytes, IRS-1/PI3K/AKT pathway was investigated. The levels of phosphorylation of IRS-1, PI3K, and AKT in BPF-treated 3T3-L1adipocytes were analyzed and the results were shown in Fig. 6A-C. The levels of phosphorylated IRS-1, PI3K and AKT were increased significantly by insulin, which means insulin activated the IRS-1/PI3K/AKT pathway. After exposure to BPF, the levels of phosphorylated IRS-1, PI3K and AKT in response to insulin stimulation were all reduced along with the increase in the concentration of BPF.

To further verify the inhibitory effect of BPF on the activation of IRS-1/PI3K/AKT signaling pathway, we used LY294002 (MCE, USA), a PI3K specific inhibitor, which can inhibit the activation of PI3K-AKT signaling pathway (Hou et al., 2018). After pretreated with LY294002 for 2 h, the cells were exposed to BPF (20 μM) for 48 h and then the changes of IRS-1 / PI3K / AKT signaling pathway and GLUT4 were measured by western blot. Our results showed that both LY294002 and BPF decreased the phosphorylation level of AKT stimulated by insulin, but there seems to be no combined effect between inhibitors and BPF. What's more, BPF did not increase the phosphorylation level of AKT without the insulin stimulation (Fig. 6D). Because the PI3K/AKT signaling pathway regulates the translocation of the insulin-sensitive glucose transporter GLUT4, we detected the expression of total GLUT4 and membrane GLUT4. The results showed that the expression of membrane GLUT4 was decreased significantly in BPF group, LY294002 group and LY294002 + BPF group. The change of membrane GLUT4 was consistent with the changes of AKT (Fig. 6E and F). However, the expression of total GLUT4 was decreased significantly only in LY294002 + BPF group. Given all of that, these results indicated that the activation of IRS-1/PI3K/AKT signaling pathway by insulin stimulation was inhibited by BPF and thus reduced GLUT4 membrane translocation in 3T3-L1 adipocytes.

4. Discussion

In this study, we investigated the effects of BPF exposure on the glucose metabolism and insulin sensitivity in 3T3-L1 adipocytes. Firstly, the results of this study showed that in 3T3-L1 adipocytes treated with BPF for 48 h the level of lipids was unchanged, probably implying the absence of lipid accumulation. Secondly, exposure of the cells to BPF for 48 h resulted in reduced levels of adipokines like *leptin, adiponectin* and *apelin* and proinflammatory cytokines, while the gene expression of *resistin* and *chemerin* was increased. Furthermore, we found that BPF reduced glucose consumption and GLUT4 translocation in insulin treated 3T3-L1 adipocytes. Finally, BPF showed to inhibit the activation

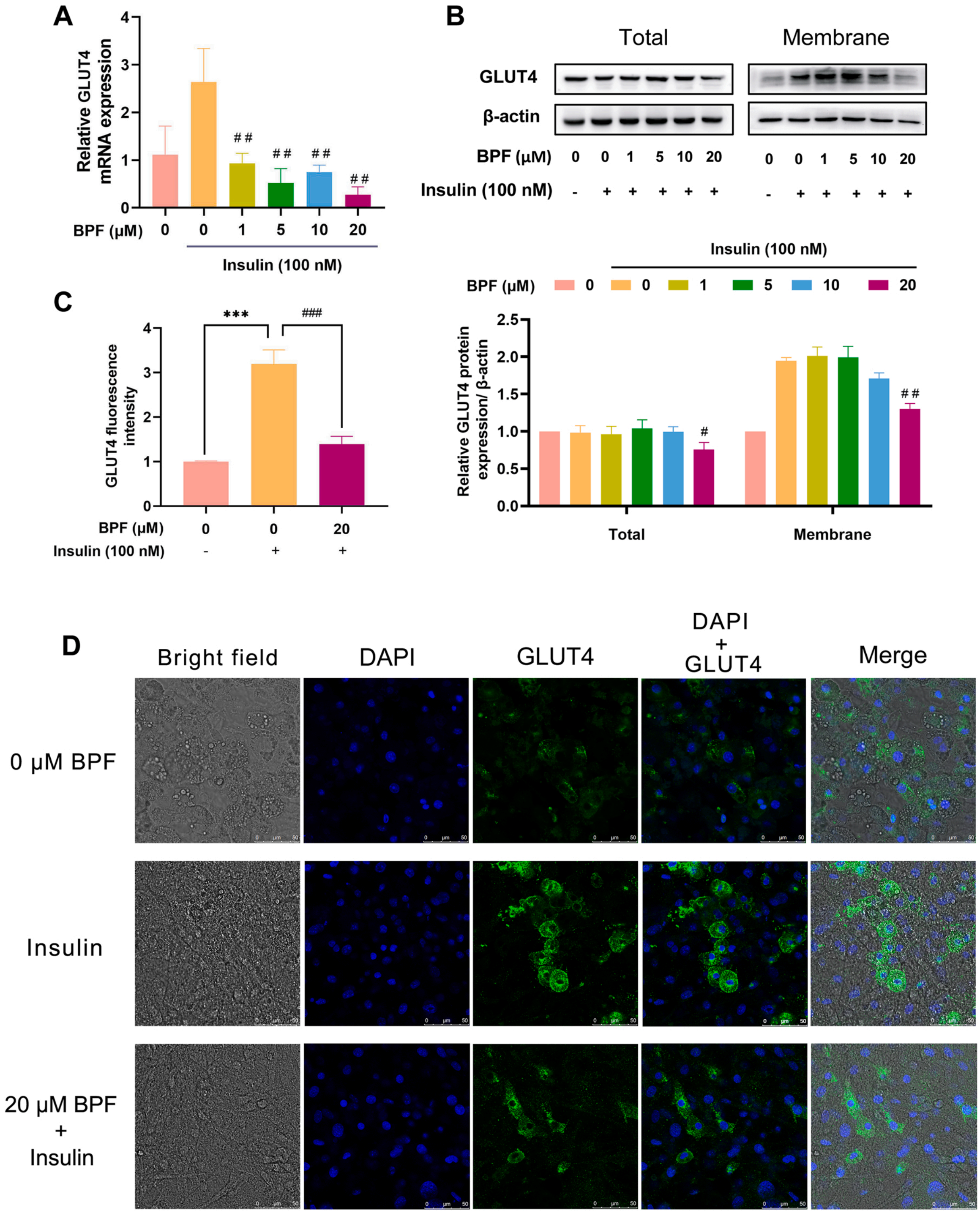

Fig. 5. BPF suppresses the insulin-stimulated glucose metabolism in 3T3-L1 adipocytes by inhibiting translocation of GLUT4. (A) the mRNA expression of GLUT4; (B) the protein expression of total and membrane GLUT4; (C) bar graph of the GLUT4 fluorescence intensity; (D) immunocytochemical imaging by confocal microscope. Data are presented as mean ± SD (n = 3). $^{***}p < 0.001$ vs. 0, $^{\#}p < 0.05$ vs. 0 + insulin stimulated, $^{\#\#}p < 0.01$ vs. 0 + insulin stimulated.

of the IRS-1/PI3K/AKT pathway, which plays an important role in insulin signal transduction.

Adipocytes are the main component of adipose tissue and are related to many physiologically and pathologically important metabolic events (Shin et al., 2020). There are two types of proliferation of adipose tissues. One is hypertrophic growth, which is attributed to the enlargement of individual adipocytes through lipid accumulation, the other is hyperplastic growth, which is the result of an increase in the number of adipocytes (Shin et al., 2020; Wang et al., 2013). Neonatal adipose tissue expands through hypertrophy and hyperplasia but the inguinal subcutaneous white adipose tissue (iWAT) in adult males is almost exclusively expanded through hypertrophy. This hypertrophic growth is highly correlated to metabolic disease (Ying and Simmons, 2020). In this study, we focused on the hypertrophy of mature adipocytes instead of the hyperplasia. However, there seemed to be no significant change in the lipid content in the 3T3-L1 adipocytes, which was consistent with some epidemiological results, which suggest that the levels of BPS and BPF were not associated with obesity in a cross-sectional study of adults adjusted for lifestyle and socio-economic factors (Liu et al., 2017).

In addition to storing energy, adipose tissue also transmits

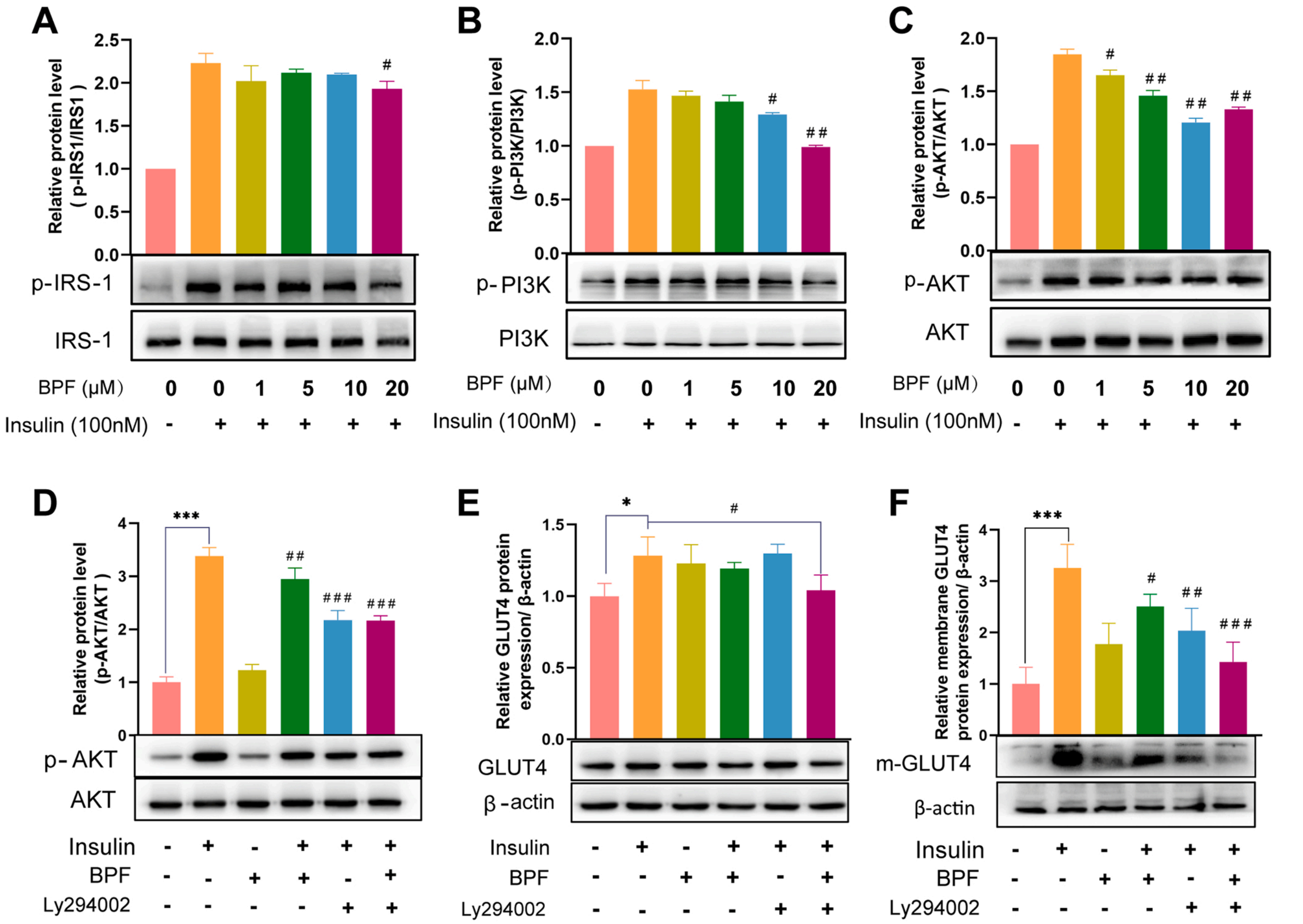

Fig. 6. BPF inhibits the IRS-1/PI3K/AKT pathway in insulin-treated 3T3-L1 adipocytes. (A-C) The phosphorylation ratio of IRS-1 (A), PI3K (B) and AKT (C) in the 3T3-L1 adipocytes in presence of insulin (100 nM) after BPF exposure for 48 h were detected by western blot. (D-E) 3T3-L1 adipocytes were treated with LY294002 (10 μM, PI3K inhibitor) for 2 h, then exposed to BPF (20 μM) for 48 h. The phosphorylation ratio of AKT (D), total GLUT4 (E) and membrane GLUT4 (F) in the 3T3-L1 adipocytes in presence of insulin (100 nM) were detected by western blots. Data are presented as mean ± SD (n = 3). * $p < 0.05$ vs. 0, * ** $p < 0.001$ vs. 0, $^{\#}p < 0.05$ vs. 0 + insulin stimulated, $^{\#\#}p < 0.01$ vs. 0 + insulin stimulated, $^{\#\#\#}p < 0.001$ vs. 0 + insulin stimulated.

information to other metabolically active organs, such as the muscles, liver, pancreas and brain, so as to regulate systemic metabolism (Parimisetty et al., 2016; Rosen and Spiegelman, 2014; Scherer, 2006). Those cytokines secreted by adipose tissues are defined as adipokines. As the first adipokine leptin was discovered in 1994 (Zhang et al., 1994), and adiponectin was cloned in 1995 (Scherer et al., 1995), other adipokines such as resistin, chemerin, apelin, visfatin, monocyte chemoattractant protein 1 (MCP-1), tumor necrosis factor α (TNF-α), interleukin 6 (IL-6), and plasminogen activator inhibitor 1 (PAI1) were found thereafter (Fasshauer and Blüher, 2015). Leptin and adiponectin are very important for regulating energy homeostasis, which are mainly secreted by adipocytes (Blüher and Mantzoros, 2015). Resistin is described as an adipocyte hormone associated with obesity and insulin resistance (Shuldiner et al., 2001). Chemerin is considered to regulate adipocyte differentiation and lipolysis (Goralski et al., 2007). The imbalance in the production or secretion of individual adipokines is related to inflammation and insulin resistance, which often occurs after obesity. A large amount of evidence shows that the impaired biosynthesis, assembly, secretion and signal transduction of adipokines may lead to adipose tissue dysfunction, which is related to the development of obesity and obesity related diseases (Deng and Scherer, 2010). For example, the association of insulin resistance, which result in T2D, with alterations in adipokines (leptin, adiponectin, resistin and so on) has been confirmed (Antoaneta et al., 2018; Gateva et al., 2018; Rabe et al., 2008; Sitar-Taut et al., 2020). Here we investigated that the effect of BPF on the expression of adipokines in 3T3-L1 adipocytes by using qRT-PCR assay. The results of this study showed that exposure to BPF for 24 h only reduced the transcription of *adiponectin, apelin* and *resistin*. However, after exposure of the cells to BPF for 48 h reduced the levels of adipokines, like *leptin, adiponectin* and *apelin* and proinflammatory cytokines were observed, while the gene expression of *resistin* and *chemerin* was increased. These changes in adipokines suggest that BPF exposure for 48 h may influence the response of the 3T3-L1 adipocytes to insulin. Therefore, we performed additional experiments to explore the potential effect of BPF in 3T3-L1 adipocytes stimulated by insulin.

In the adipocyte, one of the main functions of insulin is stimulating glucose uptake and utilization (Rahman et al., 2021). Here we focused on whether BPF impairs the glucose metabolism of 3T3-L1 adipocytes by glucose consumption assay. Consistent with our conjecture, BPF did reduce the basal glucose consumption or the insulin-stimulated glucose consumption of cells. What's more, the insulin-stimulated glucose consumption was more seriously affected by BPF. As insulin increases the glucose uptake of adipocytes by regulating the localization of GLUT4 from cytoplasm to plasma membrane (Rea and James, 1997) and the amount of GLUT4 in adipose tissue represents a good marker of systemic insulin sensitivity (Graham et al., 2006), we further investigated whether BPF suppresses the insulin-stimulated glucose consumption though GLUT4 in 3T3-L1 adipocytes. In the present study, BPF inhibited the gene and protein expression of GLUT4 and its translocation to the plasma membrane that was stimulated by insulin. These results indicate that BPF may reduce insulin-induced glucose uptake activity by interfering with GLUT4′s expression and translocation.

In Mammals, IRS have four subtypes and the roles of IRS1 and IRS2 in insulin signaling pathway and glucose metabolism have been fully confirmed (Haeusler et al., 2018). Reduction of IRS1 has been reported to be involved in insulin resistance (James et al., 2021). Antisense knockdown of IRS-1 impairs insulin-stimulated glucose uptake seriously in primary rat adipocytes (Quon et al., 1994). The importance of PI3K activation in insulin's action on adipocyte was emphasized by the study with induced deletion of PIP3 phosphatase in mouse mature adipocytes (Morley et al., 2015). The synthesis or photogenetic activation of PI3K/AKT was enough to increase the glucose uptake by adipocytes, where AKT also promoted GLUT4 translocation in adipocytes by phosphorylating target molecules involved in vesicle binding, docking and fusion (Xu et al., 2016; Yoshizaki et al., 2007). In our study, the decrease in the activation of the insulin signaling pathway, which was observed by downregulating the phosphorylation of IRS-1, PI3K and AKT, may lead to the attenuation of GLUT4 translocation.

Adipose tissue plays an important role in many metabolic pathways, including insulin signaling and insulin sensitivity, thus any agent that regulates adipocyte metabolism may in turn affect the metabolism of glucose and its homeostasis (Yaribeygi et al., 2019). In summary, this study demonstrates that BPF may suppress insulin-stimulated glucose metabolism in adipocytes through mechanisms involving the disruption of the IRS-1/PI3K/AKT signaling pathway.

5. Conclusion

BPF exposure may damage the action of insulin on glucose metabolism in adipocytes by altering the expression of some adipokines and reducing the insulin sensitivity of adipocytes through inhibiting the IRS-1/PI3K/AKT signaling pathway.

CRediT authorship contribution statement

Ming Shi: Conceptualization, Resources, Writing – review & editing, Supervision Project administration, Funding acquisition, Supervision. **Li Li**: Conceptualization, Writing – review & editing, Supervision Project administration, Funding acquisition, Supervision. **Huiling Chen**: Conceptualization, Project administration, Methodology, Software, Validation, Formal analysis, Writing – original draft, Investigation, Data curation, Visualization. **Jiangbin Li**: Conceptualization, Project administration, Methodology, Software, Validation, Formal analysis, Writing – original draft, Investigation, Data curation, Visualization. **Yanchao Zhang**: Conceptualization, Project administration, Validation, Investigation. **Wei Zhang**: Investigation, Methodology, Validation. **Xing Li**: Investigation, Methodology, Validation. **Huanwen Tang**: Project administration, Writing – review & editing**. Yungang Liu:** Project administration, Writing – review & editing**. Tianlan Li**: Investigation, Methodology. **Haoqi He**: Investigation, Methodology. **Bohai Du**: Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.113201.

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